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### Anti-fog sheet structure and helmet

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#### Abstract

An anti-fog sheet structure includes: a substrate made of polycarbonate, a first synthetic polymer film formed on a first main surface side of the substrate, and a second synthetic polymer film formed on a second main surface side, wherein the first synthetic polymer film has, on its surface, a plurality of first bumps having an equivalent circular area diameter  $Dp1$  in the range of more than 50 nm and less than 500 nm, the second synthetic polymer film has, on its surface, a plurality of second bumps having an equivalent circular area diameter  $Dp2$  in the range of more than 20 nm and less than 500 nm, and the first synthetic polymer film is formed from a cured product of a photocurable resin containing 30 mass % or more of ethoxylated pentaerythritol tetraacrylate.

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**Inventors:** Nakamatsu; Kenichiro (Kameyama, JP), Sugine; Mitsuaki (Kameyama, JP), Hashii; Hideya (Kameyama, JP), Taguchi; Tokio (Kameyama, JP)

**Applicant:** Sharp Display Technology Corporation (Kameyama, JP)

**Family ID:** 1000008818703

**Assignee:** Sharp Display Technology Corporation (Kameyama, JP)

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*Primary Examiner:* Muromoto, Jr.; Robert H

*Attorney, Agent or Firm:* ScienBiziP, P.C.

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## Background/Summary

### BACKGROUND

#### 1. Field

(1) The present disclosure relates to anti-fog sheet structures and helmets.

#### 2. Description of the Related Art

(2) For example, anti-fog face shields have been developed for helmets.

- (3) Japanese Registered Utility Model No. 3202173 discloses an anti-fog film structure used for helmets. The anti-fog film structure disclosed in Japanese Registered Utility Model No. 3202173 includes an anti-fog film produced by improving the hydrophilicity of the surface of a triacetate cellulose film by way of saponification.
- (4) International Publication No. 2012/086552 discloses an anti-fog coating resin composition including a water-absorbent resin composition, and a plastic laminate produced by using the resin composition. The anti-fog coating resin composition contains a specific (meth)acrylamide copolymer, which is a copolymer of a (meth)acrylamide monomer and another monomer, and at least one bifunctional or higher (meth)acrylate compound. The anti-fog coating resin composition is applied to a polycarbonate plate and cured by exposure to UV light to form a coated surface, and the coated surface has anti-fog properties, as described in International Publication No. 2012/086552.
- (5) Japanese Unexamined Patent Application Publication No. 2016-109979 discloses an anti-reflection plate with anti-glare properties used for information display panels placed for advertising purposes in locations where many people can see the panels. The anti-reflection plate disclosed in Japanese Unexamined Patent Application Publication No. 2016-109979 has, on each side, an anti-reflection film having a moth-eye structure on its surface. When the anti-reflection film is made of hydrophilic resin, the surface having the moth-eye structure exhibits super-hydrophilic properties and has a contact angle to water of 20° or less, as described in Japanese Unexamined Patent Application Publication No. 2016-109979. Water droplets, when attached to the super-hydrophilic surface, spread thinly and evaporate quickly. The super-hydrophilic surface thus has an anti-fog action, as described in Japanese Unexamined Patent Application Publication No. 2016-109979.
- (6) The anti-fog properties required for the face shield of a helmet may be such that, for example, the anti-fog time (the time for the transmittance to decrease to 80% of its initial value with the face shield in contact with saturated water vapor at 50° C.) in the anti-fog test in accordance with the ECE R22-5 standard is at least 20 seconds or longer, and may be 90 seconds or longer.
- (7) The structures described in Japanese Registered Utility Model No. 3202173, International Publication No. 2012/086552, and Japanese Unexamined Patent Application Publication No. 2016-109979, however, fail to achieve the anti-fog properties required for the face shield of a helmet. For example, International Publication No. 2012/086552 only evaluates exhaled breath anti-fog properties (whether fogging occurs after exhaled breath is blown to the coated surface for 5 seconds) and hot-water anti-fog properties (whether fogging occurs 60 seconds after the coated surface is placed on a beaker containing 40° C. hot water such that the coated surface faces the hot water surface). Using only the hydrophilic surface as in Japanese Registered Utility Model No. 3202173 or Japanese Unexamined Patent Application Publication No. 2016-109979 has limitations in anti-fog properties under conditions where evaporation is difficult to occur.
- (8) It is desirable to provide, for example, an anti-fog sheet structure that can satisfy an anti-fog time of 90 seconds or longer in the anti-fog test in accordance with the ECE R22-5 standard, and a helmet including the anti-fog sheet structure.

## SUMMARY

(9) According to an aspect of the disclosure, there is provided an anti-fog sheet structure including: a substrate made of a transparent plastic, a first synthetic polymer film formed on a first main surface side of the substrate, and a second synthetic polymer film formed on a second main surface side of the substrate on an opposite side from a first main surface, wherein the first synthetic polymer film has, on its surface, a plurality of first bumps having an equivalent circular area diameter  $Dp1$  in a range of more than 50 nm and less than 500 nm as viewed in a normal direction of the first synthetic polymer film, the second synthetic polymer film has, on its surface, a plurality of second bumps having an equivalent circular area diameter  $Dp2$  in a range of more than 20 nm and less than 500 nm as viewed in a normal direction of the second synthetic polymer film, the first synthetic polymer film is formed from a cured product of a photocurable resin containing 30 mass

% or more of ethoxylated pentaerythritol tetraacrylate, the second synthetic polymer film is formed from a cured product of a photocurable resin, and the anti-fog sheet structure has anti-fog properties such that a time for a visible light in-line transmittance of the first synthetic polymer film to decrease to 80% of an initial value with the surface of the first synthetic polymer film in contact with saturated water vapor at 50° C. is 90 seconds or longer.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic side view of a full-face helmet including an anti-fog sheet structure according to an embodiment of the present disclosure;
- (2) FIG. 2 is a schematic perspective view of a shield of the full-face helmet;
- (3) FIG. 3 is a schematic perspective view of an anti-fog sheet structure to be attached to the shield;
- (4) FIG. 4 is a top view of the anti-fog sheet structure attached to the shield;
- (5) FIG. 5 is a schematic horizontal sectional view of the anti-fog sheet structure attached to the shield;
- (6) FIG. 6 is a schematic sectional view of an anti-fog sheet structure according to an embodiment of the present disclosure;
- (7) FIG. 7 is a schematic sectional view of an anti-fog sheet structure according to an embodiment of the present disclosure;
- (8) FIG. 8 is an SEM image of one moth-eye structure of a first synthetic polymer film; and
- (9) FIG. 9 is an SEM image of another moth-eye structure of the first synthetic polymer film.

### DESCRIPTION OF THE EMBODIMENTS

(10) An anti-fog sheet structure and a full-face helmet including an anti-fog sheet structure according to embodiments of the present disclosure will be described below with reference to the drawings. The anti-fog sheet structure and the helmet according to the embodiments of the present disclosure are not limited to those illustrated below, and the anti-fog sheet structure according to the embodiment of the present disclosure may also be attached to helmets other than full-face helmets.

(11) First, the structure of a full-face helmet **100** including an anti-fog sheet structure **20** will be described with reference to FIGS. 1 to 5. FIG. 1 is a schematic side view of the full-face helmet **100** including an anti-fog sheet structure according to an embodiment of the present disclosure, and FIG. 2 is a schematic perspective view of a shield **10** of the full-face helmet **100**. FIG. 3 is a schematic perspective view of the anti-fog sheet structure **20** to be attached to the shield **10**. FIG. 4 is a top view of the anti-fog sheet structure **20** attached to the shield **10**, and FIG. 5 is a schematic horizontal sectional view of the anti-fog sheet structure **20** attached to the shield **10**.

(12) Referring to FIG. 1, the helmet **100** includes the shield **10** and a shell **40**. The helmet **100** further includes the anti-fog sheet structure **20** inside the shield **10** with a gap **30a** (see FIG. 5) therebetween. Referring to FIGS. 1 and 2, the shield **10** is attached to the shell **40** by inserting pins on the shell **40** into holes **10a** and **10b**. The anti-fog sheet structure **20** is removably adhered to the inner surface of the shield **10** with a seal **30** by fitting pins **12a** and **12b** of the shield **10** into notches **22a** and **22b** (see FIGS. 3 and 4). The seal **30** is made of, for example, silicone sealant and has a width of about 1 mm or more and about 5 mm or less and a height of about 1 mm or more and about 5 mm or less. The seal **30** allows a sealed gap (space) **30a** to be formed between the inner surface of the shield **10** and an outer surface **20SO** of the anti-fog sheet structure **20**. The gap **30a** has an effect of thermally insulating the anti-fog sheet structure **20** from the shield **10**. In other words, even if the shield **10** is exposed to cold outside air, the temperature of the anti-fog sheet structure **20** is less likely to drop rapidly. The shield **10** is made of, for example, polycarbonate or acrylic resin with a thickness of about 0.5 mm or more and about 10 mm or less.

(13) The helmet **100** further includes a shock absorbing liner, an interior structure, a chinstrap, and other elements, which are not shown and omitted here for simplicity. The elements of the helmet **100** other than the anti-fog sheet structure **20** may be the same as those of known full-face helmets.

(14) Referring now to FIGS. **6** and **7**, examples of the anti-fog sheet structure according to the embodiments of the present disclosure will be described.

(15) FIG. **6** illustrates the schematic sectional view of an anti-fog sheet structure **20A** according to the embodiment of the present disclosure. The anti-fog sheet structure **20A** has a substrate **22A** made of polycarbonate, a first synthetic polymer film **24A** formed on a first main surface **22AS1** side of the substrate **22A**, and a second synthetic polymer film **26A** formed on a second main surface **22AS2** side of the substrate **22A** on the opposite side from the first main surface **22AS1**. The anti-fog sheet structure **20A** is configured such that the first synthetic polymer film **24A** comes inside when the helmet is worn. In other words, the seal **30** is attached to the second synthetic polymer film **26A** formed opposite the first synthetic polymer film **24A** across the substrate **22A**, and the anti-fog sheet structure **20A** is removably adhered to the inner surface of the shield **10** with the seal **30** such that the second synthetic polymer film **26A** faces the inner surface of the shield **10**. The same applies to an anti-fog sheet structure **20B** described below. The substrate **22A** has, for example, a thickness of about 0.3 mm or more and about 1.0 mm or less. The material of the substrate **22A** may be, but not necessarily, polycarbonate used as an example from the viewpoint of the close contact between the substrate **22A** and the first synthetic polymer film **24A** and/or the close contact between the substrate **22A** and the second synthetic polymer film **26A**, or may be a transparent (visible light transmittance of 90% or more) plastic, such as acrylic resin or PET. The surface of the substrate **22A** may be treated with a primer as appropriate.

(16) The first synthetic polymer film **24A** has multiple first bumps **24Ap** on its surface, and the second synthetic polymer film **26A** has multiple second bumps **26Ap** on its surface. The multiple first bumps **24Ap** and the multiple second bumps **26Ap** form a moth-eye structure.

(17) The moth-eye structure is based on the principle of a so-called moth-eye (moth's eye). The moth-eye structure has fine bumps controlled to be smaller than or equal to the wavelength of visible light ( $\lambda=380$  nm to 780 nm) and continuously changes the refractive index for incident light from the refractive index of the incident medium to the refractive index of the substrate in the depth direction of the bumps to reduce reflection. The moth-eye structure is characterized by its ability to exhibit an anti-reflection action with small incident angle dependence over a wide wavelength range.

(18) The applicant has developed a method for producing an anti-reflection film (anti-reflection surface) with a moth-eye structure by using an anodized porous alumina layer. The use of the anodized porous alumina layer enables production of molds with an inverted moth-eye structure with high mass productivity (Japanese Unexamined Patent Application Publication No. 2009-166502, International Publication No. WO 2011/125486, and International Publication No. WO 2013/183576). All of the disclosures of Japanese Unexamined Patent Application Publication No. 2009-166502, International Publication No. WO 2011/125486, and International Publication No. WO 2013/183576 are incorporated herein by reference in their entirety.

(19) The moth-eye structure is formed by exposing, to UV light, a UV-curable resin between a substrate and a mold (moth eye-forming mold) with an inverted moth-eye structure formed by the above anodized porous alumina layer to cure the UV-curable resin so that the surface of the resulting UV-cured resin has a moth-eye structure transferred from the inverted moth-eye structure. In other words, a synthetic polymer film formed by a UV-cured resin with a moth-eye structure on its surface is obtained.

(20) As viewed in the normal direction of the first synthetic polymer film **24A**, the equivalent circular area diameter  $Dp1$  of the multiple first bumps **24Ap** is in the range of more than 50 nm and less than 500 nm. For example, when the first bumps **24Ap** are conical, the equivalent circular area diameter  $Dp1$  of the first bumps **24Ap** corresponds to the diameter of the bases of the cones. The

typical inter-distance  $D_{1int}$  between adjacent first bumps **24Ap** is more than 50 nm and 1000 nm or less. When the first bumps **24Ap** are densely arranged and no space is present between adjacent first bumps **24Ap** (e.g., the bases of the cones partially overlap each other) as illustrated in FIG. 6, the equivalent circular area diameter  $D_{p1}$  of the bumps **24Ap** is equal to the inter-distance  $D_{1int}$  between adjacent first bumps **24Ap**. The typical height  $D_{h1}$  of the first bumps **24Ap** is 50 nm or more and less than 500 nm. The thickness  $t_{sub.S1}$  of the first synthetic polymer film **24A** is not limited as long as the thickness  $t_{sub.S1}$  is larger than the height  $D_{h1}$  of the first bumps **24Ap**.

(21) As viewed in the normal direction of the second synthetic polymer film **26A**, the equivalent circular area diameter  $D_{p2}$  of the multiple second bumps **26Ap** is in the range of more than 20 nm and less than 500 nm. For example, when the second bumps **26Ap** are conical, the equivalent circular area diameter  $D_{p2}$  of the second bumps **26Ap** corresponds to the diameter of the bases of the cones. The typical inter-distance  $D_{2int}$  between adjacent second bumps **26Ap** is more than 20 nm and 1000 nm or less. When the second bumps **26Ap** are densely arranged and no space is present between adjacent second bumps **26Ap** (e.g., the bases of the cones partially overlap each other) as illustrated in FIG. 6, the equivalent circular area diameter  $D_{p2}$  of the bumps **26Ap** is equal to the inter-distance  $D_{2int}$  between adjacent second bumps **26Ap**. The typical height  $D_{h2}$  of the second bumps **26Ap** is 50 nm or more and less than 500 nm. The thickness  $t_{sub.S2}$  of the second synthetic polymer film **26A** is not limited as long as the thickness  $t_{sub.S2}$  is larger than the height  $D_{h2}$  of the second bumps **26Ap**.

(22) The first bumps **24Ap** and the second bumps **26Ap** are, for example, conical as illustrated in the figure. The first bumps **24Ap** and the second bumps **26Ap** may be bumps each having a portion with a uniform cross-sectional area (the cross section parallel to the film surface) as well as bumps each having a shape in which the cross-sectional area (the cross-section parallel to the film surface) increases toward the substrate **22A**. The two-dimensional arrangement of the multiple first bumps **24Ap** or the multiple second bumps **26Ap**, which form the moth-eye structure, on the surface may be less regular (periodic), or random.

(23) A moth eye-forming mold for forming a synthetic polymer film having multiple bumps that satisfy the relationship of  $D_{int} > D_p$  can also be produced by adjusting the time for etching the anodized porous alumina layer in the above method for producing a moth eye-forming mold. A moth-eye structure in which multiple bumps ( $D_{p1}$ ) each having a columnar shape with a conical tip are arranged at intervals of  $D_{int}$  ( $> D_p$ ) is formed.

(24) The anti-fog sheet structure **20A** includes the second synthetic polymer film **26A** in addition to the first synthetic polymer film **24A** and has a moth-eye structure on each side. This configuration prevents or reduces reflection on each side of the anti-fog sheet structure **20A**, so that the anti-fog sheet structure **20A** has ultra-low reflection performance. Since the moth-eye structure has low incidence angle dependence, the moth-eye structure has a high anti-reflection effect even when it is attached to the inner surface of the shield **10** to form a curved surface.

(25) FIG. 7 illustrates the schematic sectional view of the anti-fog sheet structure **20B** according to an embodiment of the present disclosure. The anti-fog sheet structure **20B** has a substrate **22B** made of polycarbonate, a first synthetic polymer film **24B** formed on a first main surface **22BS1** side of the substrate **22B**, a second synthetic polymer film **26B** formed on a second main surface **22BS2** side of the substrate **22B** on the opposite side from the first main surface **22BS1**, and a third synthetic polymer film **28B** formed between the first main surface **22BS1** of the substrate **22B** and the first synthetic polymer film **24B**.

(26) The first synthetic polymer film **24B** has multiple first bumps **24Bp** on its surface, the second synthetic polymer film **26B** has multiple second bumps **26Bp** on its surface, and the third synthetic polymer film **28B** has multiple third bumps **28Bp** on its surface. The multiple first bumps **24Bp**, the multiple second bumps **26Bp**, and the multiple third bumps **28Bp** form a moth-eye structure. The anti-fog sheet structure **20B** differs from the anti-fog sheet structure **20A** illustrated in FIG. 6 in having the third synthetic polymer film **28B**. The other configuration of the anti-fog sheet structure

**20B** may be the same as that of the anti-fog sheet structure **20A**. The equivalent circular area diameter **Dp3** and height **Dh3** of the multiple third bumps **28Bp** of the third synthetic polymer film **28B** may be the same as the equivalent circular area diameter **Dp2** and height **Dh2** of the multiple second bumps **26Bp** of the second synthetic polymer film **26B**. The third synthetic polymer film **28B** acts so as to improve the close contact between the first synthetic polymer film **24B** and the substrate **22B**. A fourth synthetic polymer film may be provided between the second main surface **22BS2** of the substrate **22B** and the second synthetic polymer film **26B** to improve the close contact between the second synthetic polymer film **26B** and the substrate **22B**. The fourth synthetic polymer film may be formed in the same manner as the third synthetic polymer film **28B**.

(27) When the height of the multiple first bumps **24Ap** and **24Bp** is represented by **Dh1**, **Dh1** may be 50 nm or more, and the aspect ratio  $A1=Dh1/Dp1$  may be less than 1.0. When the height of the multiple second bumps **26Ap** and **26Bp** is represented by **Dh2**, the aspect ratio  $A2=Dh2/Dp2$  may be 1.0 or more. The aspect ratio  $A2=Dh2/Dp2$  may be 1.5 or less. The height **Dh1** of the multiple first bumps **24Ap** and **24Bp** may satisfy  $150\text{ nm} \leq Dh1 \leq 300\text{ nm}$ , and the height **Dh2** of the multiple second bumps **26Ap** and **26Bp** may satisfy  $150\text{ nm} \leq Dh2 \leq 200\text{ nm}$ . The height **Dh1** of the multiple first bumps **24Ap** and **24Bp** may be greater than or equal to the height **Dh2** of the multiple second bumps **26Ap** and **26Bp**.

(28) Since the first synthetic polymer films **24A** and **24B** are water absorbent and have relatively low mechanical strength, the aspect ratio  $A1=Dh1/Dp1$  may be less than 1.0. The moth-eye structure has a larger haze value than a flat surface due to light scattering. This tendency is stronger in the first bumps **24Ap** and **24Bp**, and thus the haze value of the entire anti-fog sheet structure may be reduced by lowering the height of the second bumps **26Ap** and **26Bp**.

(29) The first synthetic polymer films **24A** and **24B** are formed from a cured product of a photocurable resin containing 30 mass % or more of ethoxylated pentaerythritol tetraacrylate, and the anti-fog sheet structures **20A** and **20B** have anti-fog properties such that the time for the visible light in-line transmittance of the first synthetic polymer films **24A** and **24B** to decrease to 80% of the initial value with the surface of the first synthetic polymer films **24A** and **24B** in contact with saturated water vapor at 50° C. is 90 seconds or longer. The second synthetic polymer films **26A** and **26B** may be formed from a hydrophilic cured product of a photocurable resin free of ethoxylated pentaerythritol tetraacrylate. The second synthetic polymer films **26A** and **26B** have, for example, hydrophilic surfaces with a (static) contact angle to water of less than 90°, as shown in Experimental Example below. The contact angle may be measured by a standard method using a contact angle meter.

(30) As described below by way of Experimental Example, the first synthetic polymer films **24A** and **24B** may be formed by using, for example, the following materials. The proportions (U-105 to U-119) of the materials used in Experimental Example are shown in Tables 1 and 2 below.

Ethoxylated pentaerythritol tetraacrylate (product name: ATM-35E, available from Shin-Nakamura Chemical Co., Ltd.) Hydrophilic urethane acrylate (product name: UA-7100, available from Shin-Nakamura Chemical Co., Ltd.) Hydrophilic urethane acrylate (product name: UA-W2A, available from Shin-Nakamura Chemical Co., Ltd.) 2-(Vinyloxyethoxy)ethyl acrylate (product name: VEEA, available from Nippon Shokubai Co. Ltd.) n-Vinyl-2-pyrrolidone (NVP, available from Nippon Shokubai Co. Ltd.) Dimethylacrylamide (product name: DMAA, available from KJ Chemicals Corporation) Phenylbis(2,4,6-trimethylbenzoyl) phosphine oxide (initiator product name: Omnirad-819, available from IGM Resins B.V.) Hydrophobic silicone additive (product name: JAR-36, available from Jujo Chemical Co., Ltd.)

(31) Ethoxylated pentaerythritol tetraacrylate (ATM-35E) contains 35 ethylene oxide (EO) groups in its molecule, is highly hydrophilic, and when added in an amount of 30 mass % or more to the photocurable resin composition (excluding the initiator and the additive), may impart water absorbency to the cured photocurable resin. The surface of the photocured resin not only has hydrophilic properties but also has hygroscopic properties as a bulk.

(32) Hydrophilic urethane acrylates (UA-7100 and UA-W2A) have high water resistance and high viscosity and thus easily form thick films. UA-7100 is less hydrophilic than UA-W2A. The water absorption rate of a cured product of a mixture of 70 mass % of UA-7100 and 30 mass % of 2-(2-vinyloxyethoxy)ethyl acrylate monomer is 2.0% or less, whereas the water absorption rate of a cured product of a mixture of 70 mass % of UA-W2A and 30 mass % of 2-(2-vinyloxyethoxy)ethyl acrylate monomer is more than 2.0%.

(33) 2-(Vinyloxyethoxy)ethyl acrylate (VEEA) has a small molecular weight and has the effect of improving close contact between the first synthetic polymer film and the polycarbonate substrate through its anchor effect.

(34) n-Vinyl-2-pyrrolidone (NVP) also has the effect of improving close contact between the first synthetic polymer film and the polycarbonate substrate and is more hydrophilic than 2-(vinyloxyethoxy)ethyl acrylate (VEEA).

(35) Like n-vinyl-2-pyrrolidone (NVP), dimethylacrylamide (DMAA) is highly hydrophilic and improves close contact between the first synthetic polymer film and the polycarbonate substrate.

(36) Phenylbis(2,4,6-trimethylbenzoyl) phosphine oxide (Omnirad-819) is a general-purpose UV-sensitive initiator.

(37) The hydrophobic silicone additive (JAR-36), when added in a small amount, allows water drops attached to the surface of the first synthetic polymer film **24A** to stay as water drops on the surface without water spreading, which keeps visibility. The contact angle of the surface of the first synthetic polymer film **24A** to water may be 20° or more.

(38) The second synthetic polymer films **26A** and **26B** may be formed by using, for example, the following materials. The proportions (G05) of the materials used in Experimental Example are shown in Table 1 below. The second synthetic polymer films **26A** and **26B** are provided to further improve the anti-reflection performance of the anti-fog sheet structures **20A** and **20B** and formed by using, for example, the photocurable resins disclosed in Japanese Unexamined Patent Application Publication No. 2009-166502, International Publication No. WO 2011/125486, and International Publication No. WO 2013/183576, as described above. The surfaces of the second synthetic polymer films **26A** and **26B** are not necessarily water absorbent and may be hydrophilic or hydrophobic. Polyethylene glycol diacrylate (product name: Miramer M280, available from Miwon Specialty Chemical Co.) Trimethylolpropane triacrylate (product name: Miramer M300, available from Miwon Specialty Chemical Co.) 2-(Vinyloxyethoxy)ethyl acrylate (Product name: VEEA, available from Nippon Shokubai Co. Ltd.) Phenylbis(2,4,6-trimethylbenzoyl) phosphine oxide (initiator product name: Omnirad-819, available from IGM Resins B.V.) Diglycerol monolaurate (hydrophilic antiviral agent, product name: POEM DL-100, available from Riken Vitamin Co., Ltd.)

(39) Diglycerol monolaurate may be omitted.

(40) The third synthetic polymer film **28B** may be formed by using, for example, the following materials. The proportions (BT7100, BTW2A) of the materials used in Experimental Example are shown in Table 3 below. Hydrophilic urethane acrylate (product name: UA-7100, available from Shin-Nakamura Chemical Co., Ltd.) Hydrophilic urethane acrylate (product name: UA-W2A, available from Shin-Nakamura Chemical Co., Ltd.) 2-(Vinyloxyethoxy)ethyl acrylate (product name: VEEA, available from Nippon Shokubai Co. Ltd.) Phenylbis(2,4,6-trimethylbenzoyl) phosphine oxide (initiator product name: Omnirad-819, available from IGM Resins B.V.)

(41) The photocurable resin composition (excluding the initiator) of the third synthetic polymer film **28B** contains 30 mass % of 2-(vinyloxyethoxy)ethyl acrylate (VEEA), and thus the third synthetic polymer film **28B** comes in very close contact with the polycarbonate substrate **22B**. The photocurable resin composition contains hydrophilic urethane acrylates (UA-7100 and UA-W2A), and thus the third synthetic polymer film **28B** also comes in very close contact with the first synthetic polymer film **24B**.

(42) Next, the anti-fog sheet structures according to the embodiments of the present disclosure will



be described by way of Experimental Example. The formulations of the photocurable resin compositions used in the experiment are shown in Table 1 to Table 3 below.

(43) TABLE-US-00001 TABLE 1 Viscosity Resin Composition Material Material Name (cp) G05 U-105 U-106 U-107 U-108 U-109 U-110 Ethoxylated pentaerythritol ATM-35E 350.0 — 60.0 60.0 65.0 50.0 50.0 50.0 tetraacrylate Urethane acrylate UA-7100 21000.0 — 30.0 25.0 20.0 30.0 30.0 20.0 UA-W2A 33400.0 — — — — — — 10.0 Polyethylene glycol M280 65.0 55.0 — — — — — diacrylate Trimethylolpropane M300 120.0 10.0 — — — — — triacrylate 2-(Vinyloxyethoxy)ethyl VEEA 3.7 35.0 10.0 15.0 15.0 8.0 — 8.0 acrylate n-Vinyl-2-pyrrolidone NVP 2.1 — — — — 12.0 20.0 12.0 Dimethylacrylamide DMAA 1.2 — — — — — — Initiator Omnirad-819 — 1.0 1.0 1.0 1.0 1.0 1.0 Diglycerol monolaurate POEM DL-100 — 1.0 — — — — — Silicone additive JAR-36 — — 0.1 0.1 0.1 0.1 0.1 0.1

(44) TABLE-US-00002 TABLE 2 Viscosity Resin Composition Material Material Name (cp) U-111 U-112 U-113 U-115 U-117 U-118 U-119 Ethoxylated ATM-35E 350.0 52.0 52.0 62.0 47.0 30.0 30.0 37.3 pentaerythritol tetraacrylate Urethane acrylate UA-7100 21000.0 30.0 30.0 — — — — — UA-W2A 33400.0 — — 25.0 35.0 45.0 40.0 38.2 Polyethylene glycol M280 65.0 — — — — — diacrylate Trimethylolpropane M300 120.0 — — — — — — triacrylate 2-(Vinyloxyethoxy)ethyl VEEA 3.7 — 5.0 13.0 18.0 25.0 30.0 6.9 acrylate n-Vinyl-2-pyrrolidone NVP 2.1 — — — — — — 17.6 Dimethylacrylamide DMAA 1.2 18.0 13.0 — — — — — Initiator Omnirad-819 — 1.0 1.0 1.0 1.0 1.0 1.0 Diglycerol monolaurate POEM DL-100 — — — — — — — Silicone additive JAR-36 — 0.1 0.1 0.1 0.1 0.1 0.1 0.05

(45) TABLE-US-00003 TABLE 3 Material Viscosity Resin Composition Material Material Name (cp) BT7100 BTW2A Urethane acrylate UA-7100 21000.0 70.0 — UA-W2A 33400.0 — 70.0 2-(Vinyloxyethoxy)ethyl VEEA 3.7 30.0 30.0 acrylate Initiator Omnirad-819 — 1.0 1.0

(46) In Experimental Example, a polycarbonate film (NF-2000 clear, available from Mitsubishi Gas Chemical Company, Inc.), 0.7 mm thick and 7 cm×7 cm, was used as a substrate.

(47) The equivalent circular area diameter Dp1 of the multiple first bumps and the inter-distance Dint1 between adjacent first bumps in the first synthetic polymer film were 150 nm and 180 nm, respectively.

(48) The equivalent circular area diameter Dp2 of the multiple second bumps and the inter-distance Dint2 between adjacent second bumps in the second synthetic polymer film were 150 nm and 150 nm, respectively. The equivalent circular area diameter Dp3 of the multiple third bumps and the inter-distance Dint3 between adjacent third bumps in the third synthetic polymer film were the same as the equivalent circular area diameter Dp2 of the second bumps and the inter-distance Dint2 between adjacent second bumps. In other words, the second synthetic polymer film and the third synthetic polymer film were formed by using the same moth eye-forming mold.

(49) FIG. 8 and FIG. 9 illustrate examples of the SEM images of the moth-eye structures of the first synthetic polymer film. The moth-eye structure in FIG. 8 had Dp1 of 150 nm, Dint1 of 150 nm, and Dh1 of 130 nm. The moth-eye structure in FIG. 9 had Dp1 of 150 nm, Dint1 of 180 nm, and Dh1 of 120 nm. The SEM images were obtained by using electron microscope S-4700 available from Hitachi High-Tech Corporation. The SEM images show that desired moth-eye structures are formed.

(50) The haze value, the total light transmittance, the reflectance (Y-value), a\* and b\* of transmitted light in the L\*a\*b\* color space, and the contact angle to water were measured for Sample A of an anti-fog sheet structure having U-108 (thickness: 130 μm)/PC substrate (thickness: 0.7 mm)/G05 (thickness: 15 μm). The haze value and the total light transmittance were measured by using turbidimeter NDH2000 available from Nippon Denshoku Industries Co., Ltd. The reflectance and the chromaticity (a\* and b\*) were measured by using spectrophotometer V-560 available from JASCO Corporation. The contact angle to water was measured by using portable contact angle meter V-560 available from Kyowa Interface Science Co., Ltd. The results are shown in Table 4 below.

(51) TABLE-US-00004 TABLE 4 Haze Total Light Reflectance Transmitted Light Contact Value Transmittance Y-Value Chromaticity Angle to (%) (%) (%) a\* b\* Water (°) 0.48 99.0 0.67 -0.32 0.69 30.6

(52) Sample A of the anti-fog sheet structure has a low haze value of 1.0(%) or less, a high total light transmittance for visible light of 90.0% or more, and a low reflectance of 1.0% or less. The absolute value of a\* and the absolute value of b\* in the L\*a\*b\* color space are each independently 1.0 or less, indicating that there is no coloration. The contact angle of the first synthetic polymer film to water is 20° or more, and water drops attached to the surface of the first synthetic polymer film stay as water drops on the surface without water spreading.

(53) The anti-fog sheet structures according to the embodiments of the present disclosure have an asymmetric structure in the thickness direction as illustrated in FIGS. 6 and 7 and, for example, when immersed in water, the anti-fog sheet structures curl such that the first synthetic polymer film side becomes convex. For example, Sample A of the anti-fog sheet structure was immersed in tap water in a stainless steel bat, left to stand for 5 hours, then pulled up to the air, and observed for its appearance. Sample A curled to a height of about 5 cm such that first synthetic polymer film side became convex.

(54) An example of evaluation of anti-fog properties using Sample B will be described below. Hot water of 50° C. (+0.5° C.) was poured into a hot bath, and Sample B was placed above the hot bath such that the first synthetic polymer film (water-absorbing layer) was in contact with saturated water vapor. The time (anti-fog time) until fogging was visually observed was measured. The visible light transmittance at the time when fogging was visually observed was higher than 80%, which can be said to be a slightly stricter condition than in the anti-fog test in accordance with the ECE R22-5 standard. The dependence of the thickness of the first synthetic polymer film (water-absorbing layer) on the anti-fog time was also evaluated. The results are shown in Table 5 below.

(55) TABLE-US-00005 TABLE 5 Film Thickness (μm) Anti-Fog First Synthetic Third Synthetic Time Polymer Film Polymer Film Total (s) 22.4 23.0 45.4 135 24.1 21.9 46.0 96 24.7 22.8 47.5 114 25.0 23.0 48.0 129 33.0 22.0 55.0 145 45.1 17.1 62.2 225 85.0 58.0 143.0 500

(56) It is revealed that the anti-fog time is 90 seconds or longer when the first synthetic polymer film has a thickness of about 20 μm. It is also revealed that the anti-fog time increases almost in proportion to the thickness of the first synthetic polymer film. From the viewpoint of mass production, the first synthetic polymer film may have a thickness not more than about 150 μm, and the anti-fog time may be 90 seconds or longer when the first synthetic polymer film has a thickness of about 20 μm or more and about 100 μm or less.

(57) The results of evaluation of the hydrophilicity and water absorbency of the first synthetic polymer film using the resin compositions (U-105 to U-119) in Tables 1 and 2 are shown in Tables 6 to 8 below. The hydrophilicity was evaluated by “observing the behavior in the evaluation of exhaled-breath and hot-water anti-fog properties (50° C. water bath surface)”, where A indicates that the film does not fog with exhaled breath and maintains the water film state above the 50° C. water bath surface, and B indicates that the film does not fog with exhaled breath and, above the 50° C. water bath surface, does not change or become a water film for a certain time but starts to fog later. The water absorbency is evaluated by “evaluation of hot-water anti-fog properties (50° C. hot bath surface) (measurement of anti-fog time),” where B indicates that the anti-fog time is 90 seconds or longer, C indicates the anti-fog time is 20 seconds or longer and shorter than 90 seconds, and D indicates that the anti-fog time is shorter than 20 seconds.

(58) After the hydrophilicity and water absorbency tests, the film may peel off or deform due to swelling (water resistance). U-113, U-115, U-117, and U-118 have some issues with water resistance. If the resin is soft, adjacent bumps in the moth-eye structure may come into close contact with each other, which is called sticking. U-111 was susceptible to sticking, depending on the height of the bumps. U-105 and U-119 were particularly superior in terms of water resistance and sticking resistance, in addition to hydrophilicity and water absorbency. The close contact

between the first synthetic polymer film and the substrate may be improved by, for example, providing an adhesive layer (third synthetic polymer film) and by selecting the material of the adhesive layer, as appropriate.

(59) Of the anti-fog sheet structure samples used in the evaluation, U-105 has the anti-fog sheet structure **20B** illustrated in FIG. 7, and the other samples have the anti-fog sheet structure **20A** illustrated in FIG. 6. The third synthetic polymer film was made of BT7100 (50 μm thick). The second synthetic polymer film was made of G05 (10 μm thick), and the first synthetic polymer film **50** was 70 μm thick.

(60) TABLE-US-00006 TABLE 6 G05 U-105 U-106 U-107 U-108 U-109 U-110 Hydrophilicity A B B B B B Water Absorbency D B C C B B B

(61) TABLE-US-00007 TABLE 7 U-111 U-112 U-113 U-115 U-117 U-118 U-119 Hydrophilicity B B B B B B Water Absorbency B B B B B B B

(62) TABLE-US-00008 TABLE 8 BT7100 BTW2A Hydrophilicity B B Water Absorbency C B

(63) The water absorption rate of the first synthetic polymer films (U-105, U-108, U-110, U-112) and the third synthetic polymer films (BT7100, BTW2A) were determined as follows.

(64) The first synthetic polymer film or the third synthetic polymer film was formed on a polycarbonate film (0.7 mm thick, 7 cm×7 cm, 4.246 g), which was a substrate. The initial mass A of each of the obtained samples was measured, and the mass B of each sample when it started to fog in the anti-fog test was measured. The mass of water absorbed by the first synthetic polymer film or the third synthetic polymer film was obtained from B-A. The ratio (expressed as percentage) of the mass of absorbed water to the initial mass of the first third synthetic polymer film or the third synthetic polymer film was calculated as a water absorption rate. The obtained results are shown in Table 9.

(65) TABLE-US-00009 TABLE 9 Average Total Mass (g) Water Mass of Water Water B. after anti-Absorption Cured Absorption Absorption No. Sample A. initial fog test B – A (g) Film (g) Rate (%) Rate (%) 1 U-105 4.667 4.712 0.045 0.421 10.570 10.583 2 4.598 4.635 0.037 0.352 10.597 3 U-108 4.605 4.661 0.056 0.359 15.535 13.817 4 4.580 4.623 0.043 0.334 12.874 5 4.568 4.610 0.042 0.322 13.043 6 U-110 4.630 4.710 0.080 0.384 20.833 17.665 7 4.598 4.658 0.060 0.352 17.045 8 4.693 4.760 0.067 0.446 15.118 9 U-112 4.711 4.767 0.056 0.465 12.043 12.106 10 4.585 4.619 0.034 0.339 10.029 11 4.611 4.663 0.052 0.365 14.247 12 BT-7100 5.211 5.222 0.011 0.965 1.161 1.532 13 5.144 5.161 0.017 0.898 1.904 14 BT-W2A 5.100 5.170 0.070 0.854 8.208 7.380 15 5.399 5.474 0.075 1.153 6.551

(66) It is revealed that all of the first synthetic polymer films have a high water absorption rate of more than 10%. The third synthetic polymer films had a low water absorption rate. In particular, the water absorption rate of BT7100 was as low as 2% or less.

(67) The abrasion resistance of anti-fog sheet structure according to embodiments of the present disclosure was evaluated. The anti-fog sheet structure illustrated in FIGS. 8 and 9 and a commercially available anti-fog sheet were used as samples. The evaluation was performed with steel wool (product name: #0000, available from Bonstar Corporation), a rubbing speed of 6000 mm/min, a rubbing distance of 10 mm, and a rubbing frequency of 10 times.

(68) Deep scratches were formed on the commercial product in the abrasion resistance test, whereas the scratches were slight on all of the anti-fog sheet structure according to the embodiments of the present disclosure. This indicates that the anti-fog sheet structure according to the embodiments of the present disclosure have sufficient abrasion resistance.

(69) The anti-fog sheet structure according to the embodiments of the present disclosure are suitably used for helmets and other products.

(70) Embodiments of the present disclosure provide solutions described in the following items.

(71) [Item 1]

(72) An anti-fog sheet structure comprising: a substrate made of a transparent plastic, a first synthetic polymer film formed on a first main surface side of the substrate, and a second synthetic

polymer film formed on a second main surface side of the substrate on an opposite side from a first main surface, wherein the first synthetic polymer film has, on its surface, a plurality of first bumps having an equivalent circular area diameter  $Dp1$  in a range of more than 50 nm and less than 500 nm as viewed in a normal direction of the first synthetic polymer film, the second synthetic polymer film has, on its surface, a plurality of second bumps having an equivalent circular area diameter  $Dp2$  in a range of more than 20 nm and less than 500 nm as viewed in a normal direction of the second synthetic polymer film, the first synthetic polymer film is formed from a cured product of a photocurable resin containing 30 mass % or more of ethoxylated pentaerythritol tetraacrylate, the second synthetic polymer film is formed from a cured product of a photocurable resin, and the anti-fog sheet structure has anti-fog properties such that a time for a visible light in-line transmittance of the first synthetic polymer film to decrease to 80% of an initial value with the surface of the first synthetic polymer film in contact with saturated water vapor at 50° C. is 90 seconds or longer. The transparent plastic forming the substrate is preferably polycarbonate.

[Item 2]

(73) The anti-fog sheet structure according to Item 1, wherein, when a height of the plurality of first bumps is represented by  $Dh1$ ,  $Dh1$  is 50 nm or more, and an aspect ratio  $A1=Dh1/Dp1$  is less than 1.0; and when a height of the plurality of second bumps is represented by  $Dh2$ , an aspect ratio  $A2=Dh2/Dp2$  is 1.0 or more.

[Item 3]

(74) The anti-fog sheet structure according to Item 2, wherein the height  $Dh1$  of the plurality of first bumps satisfies  $150\text{ nm} \leq Dh1 \leq 300\text{ nm}$ .

(75) [Item 4]

(76) The anti-fog sheet structure according to Item 2 or 3, wherein the height  $Dh2$  of the plurality of second bumps satisfies  $150\text{ nm} \leq Dh2 \leq 200\text{ nm}$ .

(77) [Item 5]

(78) The anti-fog sheet structure according to any one of Items 2 to 4, wherein the aspect ratio  $A2=Dh2/Dp2$  is 1.5 or less.

(79) [Item 6]

(80) The anti-fog sheet structure according to any one of Items 2 to 5, wherein the height  $Dh1$  of the plurality of first bumps is greater than or equal to the height  $Dh2$  of the plurality of second bumps.

(81) [Item 7]

(82) The anti-fog sheet structure according to any one of Items 1 to 6, wherein the surface of the first synthetic polymer film has a contact angle to water of 20° or more.

(83) [Item 8]

(84) The anti-fog sheet structure according to any one of Items 1 to 7, wherein the first synthetic polymer film is formed from a cured product of the photocurable resin further containing urethane acrylate.

(85) [Item 9]

(86) The anti-fog sheet structure according to Item 8, wherein the urethane acrylate is such that a cured product of a mixture of 70 mass % of the urethane acrylate and 30 mass % of 2-(2-vinyloxyethoxy)ethyl acrylate monomer has a water absorption rate of more than 2.0%.

(87) [Item 10]

(88) The anti-fog sheet structure according to any one of Items 1 to 9, further comprising a third synthetic polymer film formed between the first main surface of the substrate and the first synthetic polymer film, wherein the third synthetic polymer film has, on its surface, a plurality of third bumps having an equivalent circular area diameter  $Dp3$  in a range of more than 20 nm and less than 500 nm as viewed in a normal direction of the third synthetic polymer film.

[Item 11]

(89) The anti-fog sheet structure according to any one of Items 1 to 10, wherein the anti-fog sheet

structure has a haze value of 1.0 or less.

(90) [Item 12]

(91) The anti-fog sheet structure according to any one of Items 1 to 11, wherein an absolute value of  $a^*$  and an absolute value of  $b^*$  of transmitted light in an  $L^*a^*b^*$  color space are each independently 1.0 or less.

(92) [Item 13]

(93) A helmet comprising a shield and a shell, wherein the helmet further comprises an anti-fog sheet structure inside the shield with a gap therebetween, the anti-fog sheet structure is the anti-fog sheet structure according to any one of Items 1 to 12 and configured such that the first synthetic polymer film comes inside when the helmet is worn.

(94) The present disclosure contains subject matter related to that disclosed in Japanese Priority Patent Application JP 2023-190014 filed in the Japan Patent Office on Nov. 7, 2023, the entire contents of which are hereby incorporated by reference.

(95) It should be understood by those skilled in the art that various modifications, combinations, sub-combinations and alterations may occur depending on design requirements and other factors insofar as they are within the scope of the appended claims or the equivalents thereof.

## Claims

1. An anti-fog sheet structure comprising: a substrate made of a transparent plastic, a first synthetic polymer film formed on a first main surface side of the substrate, and a second synthetic polymer film formed on a second main surface side of the substrate on an opposite side from a first main surface, wherein the first synthetic polymer film has, on its surface, a plurality of first bumps having an equivalent circular area diameter  $Dp1$  in a range of more than 50 nm and less than 500 nm as viewed in a normal direction of the first synthetic polymer film, the second synthetic polymer film has, on its surface, a plurality of second bumps having an equivalent circular area diameter  $Dp2$  in a range of more than 20 nm and less than 500 nm as viewed in a normal direction of the second synthetic polymer film, the first synthetic polymer film is formed from a cured product of a photocurable resin containing 30 mass % or more of ethoxylated pentaerythritol tetraacrylate, the second synthetic polymer film is formed from a cured product of a photocurable resin, and the anti-fog sheet structure has anti-fog properties such that a time for a visible light in-line transmittance of the first synthetic polymer film to decrease to 80% of an initial value with the surface of the first synthetic polymer film in contact with saturated water vapor at 50° C. is 90 seconds or longer.

2. The anti-fog sheet structure according to claim 1, wherein, when a height of the plurality of first bumps is represented by  $Dh1$ ,  $Dh1$  is 50 nm or more, and an aspect ratio  $A1=Dh1/Dp1$  is less than 1.0; and when a height of the plurality of second bumps is represented by  $Dh2$ , an aspect ratio  $A2=Dh2/Dp2$  is 1.0 or more.

3. The anti-fog sheet structure according to claim 2, wherein the height  $Dh1$  of the plurality of first bumps satisfies  $150\text{ nm} \leq Dh1 \leq 300\text{ nm}$ .

4. The anti-fog sheet structure according to claim 2, wherein the height  $Dh2$  of the plurality of second bumps satisfies  $150\text{ nm} \leq Dh2 \leq 200\text{ nm}$ .

5. The anti-fog sheet structure according to claim 2, wherein the aspect ratio  $A2=Dh2/Dp2$  is 1.5 or less.

6. The anti-fog sheet structure according to claim 2, wherein the height  $Dh1$  of the plurality of first bumps is greater than or equal to the height  $Dh2$  of the plurality of second bumps.

7. The anti-fog sheet structure according to claim 1, wherein the surface of the first synthetic polymer film has a contact angle to water of 20° or more.

8. The anti-fog sheet structure according to claim 1, wherein the first synthetic polymer film is formed from a cured product of the photocurable resin further containing urethane acrylate.

9. The anti-fog sheet structure according to claim 8, wherein the urethane acrylate is such that a cured product of a mixture of 70 mass % of the urethane acrylate and 30 mass % of 2-(2-vinyloxyethoxy)ethyl acrylate monomer has a water absorption rate of more than 2.0%.
10. The anti-fog sheet structure according to claim 1, further comprising a third synthetic polymer film formed between the first main surface of the substrate and the first synthetic polymer film, wherein the third synthetic polymer film has, on its surface, a plurality of third bumps having an equivalent circular area diameter  $Dp_3$  in a range of more than 20 nm and less than 500 nm as viewed in a normal direction of the third synthetic polymer film.
11. The anti-fog sheet structure according to claim 1, wherein the anti-fog sheet structure has a haze value of 1.0 or less.
12. The anti-fog sheet structure according to claim 1, wherein an absolute value of  $a^*$  and an absolute value of  $b^*$  of transmitted light in an  $L^*a^*b^*$  color space are each independently 1.0 or less.
13. A helmet comprising a shield and a shell, wherein the helmet further comprises an anti-fog sheet structure inside the shield with a gap therebetween, the anti-fog sheet structure is the anti-fog sheet structure according to claim 1 and configured such that the first synthetic polymer film comes inside when the helmet is worn.
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